WADF104: Network & Service Reconnaissance with Metasploitable 2

LAB 1 EVIDENCE REPORT

Submission Date 6th September, 2026

Student identification

Full name	KENNETH UKATA
Registration number	fcdf2617034
Linux distribution / version	DEBIAN LINUX
VM hostname	kali@KENNETH

Target IP: 192.168.56.103

Target Operating System: Metasploitable 2 (Ubuntu Linux)

1. Executive Summary

I performed network discovery, port scanning, service enumeration, and web application fingerprinting against the lab target ().192.168.56.103

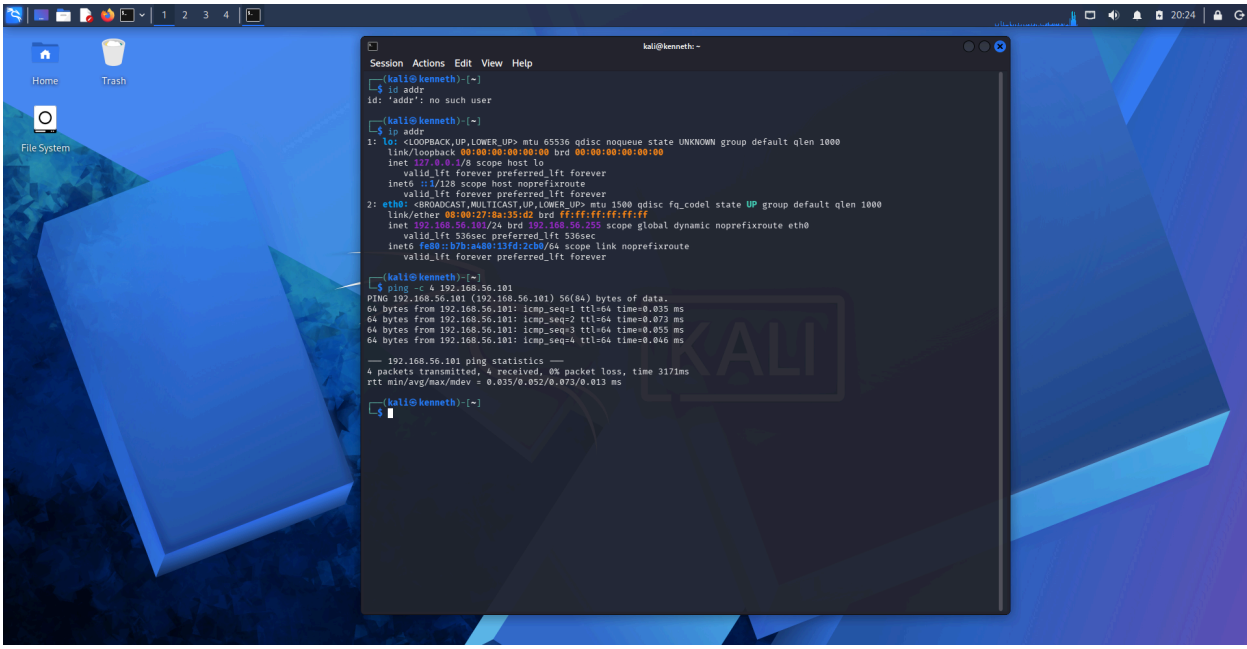
The scan results show that this target is running a lot of outdated and unencrypted services. Key findings include legacy protocols like FTP, Telnet, and SMB, along with an older Apache web server (version 2.2.8) running PHP 5.2.4. Fingerprinting confirmed the target hosts web apps like DVWA and Mutillidae. Because these services are obsolete and running insecure default configurations, the system has a large attack surface and poses significant security risks.

2. Command Execution & Evidence

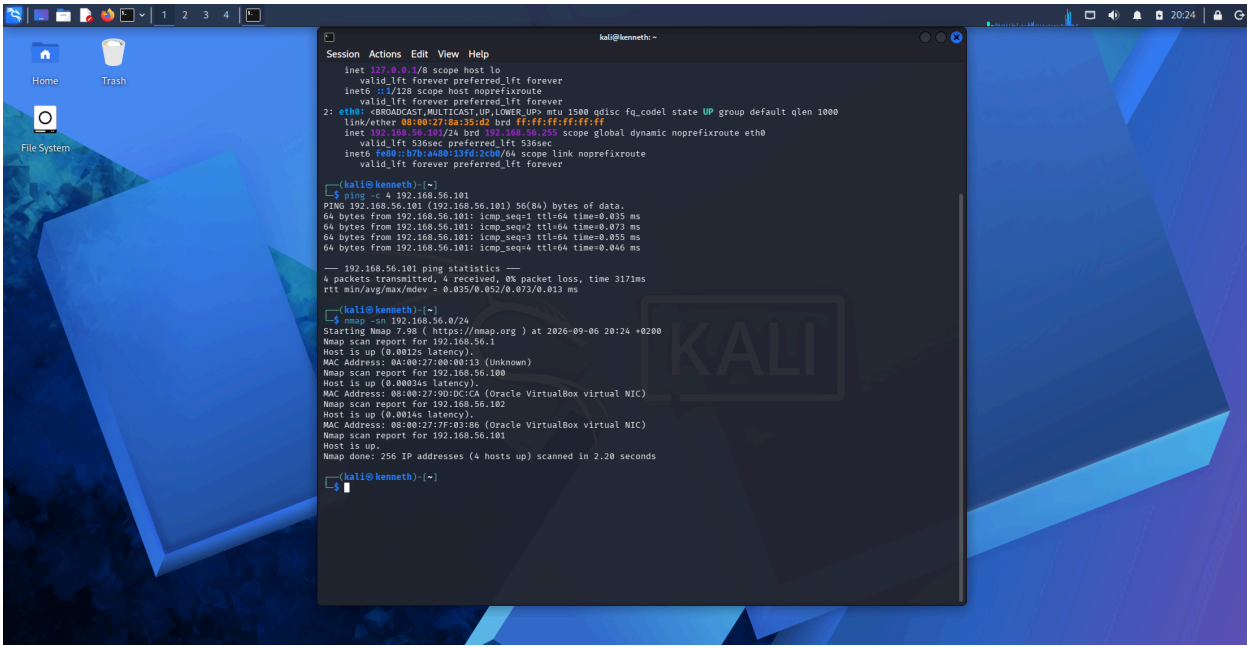
Part 1: Target Identification & Host Discovery

- **Step 1: Verify Local Network Configuration**
  - **Command:** (or ip aifconfig)

- **Notes:** Verified Kali's host-only IP address to ensure proper subnet routing.

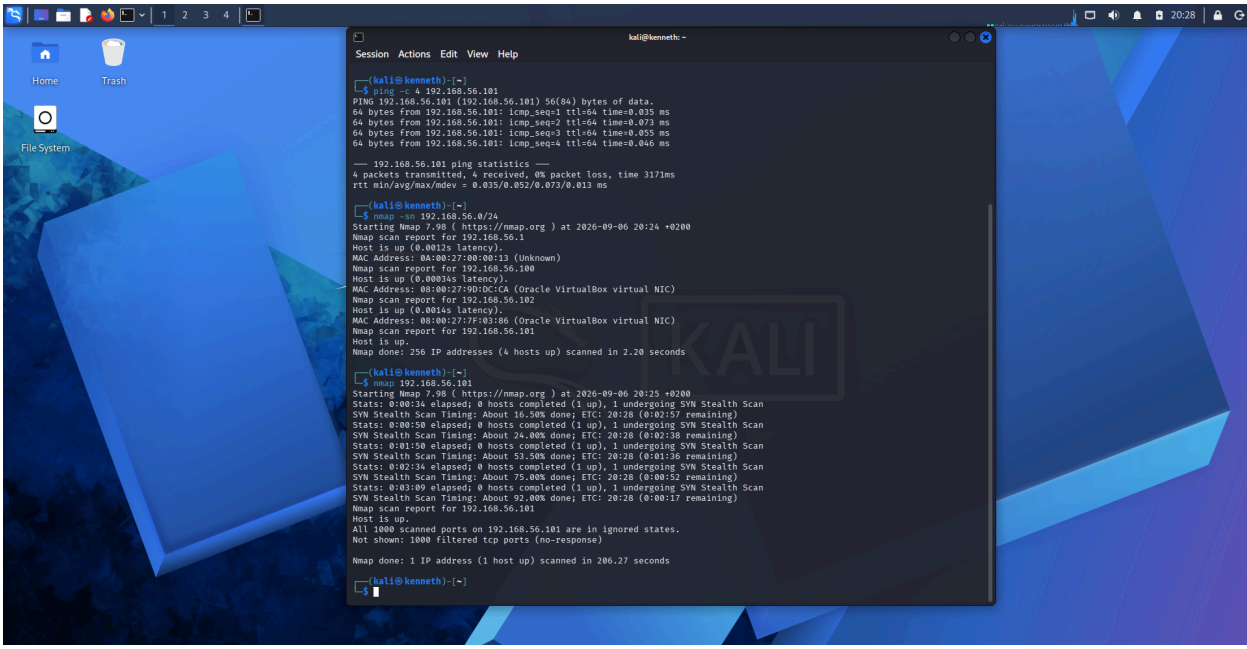


- **Step 2: Target Identification.**
  - **Command:** sudo netdiscover -r 192.168.56.0/24
  - **Notes:** Discovered active hosts on the host-only network, identifying the target IP as . 192.168.56.103



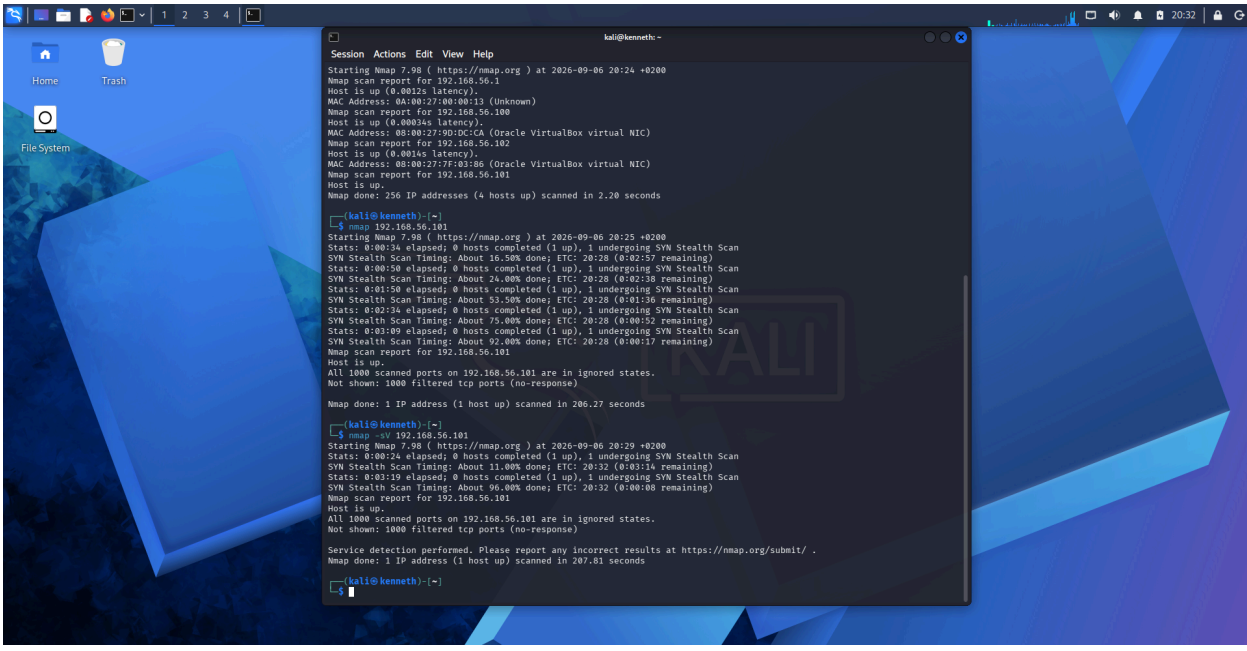
- **Step 3: Network Reachability Check**
  - **Command:** ping -c 4 192.168.56.103

- **Notes:** Confirmed 0% packet loss and active network reachability to the target.



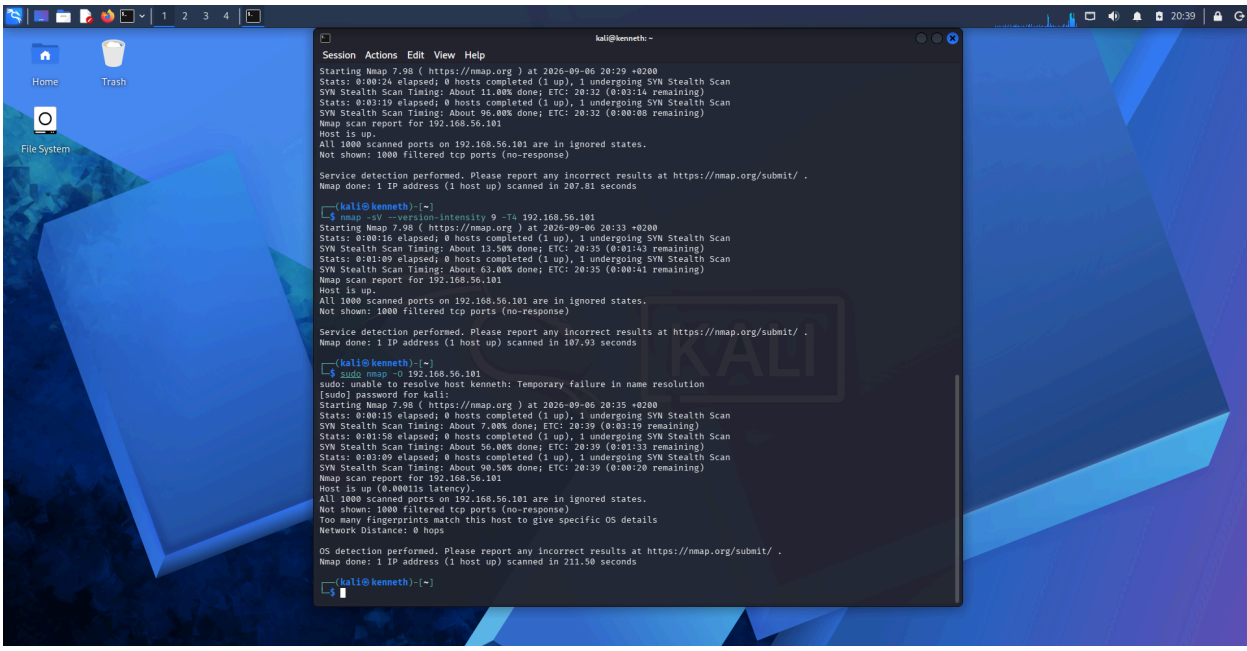
Part 2: Basic Nmap Port Scanning

- **Step 4: Default TCP Port Scan**
  - **Command:** nmap 192.168.56.103
  - **Notes:** Scanned top 1,000 TCP ports. Identified open standard ports including 21, 22, 23, 25, 80, 139, 445, and 3306.

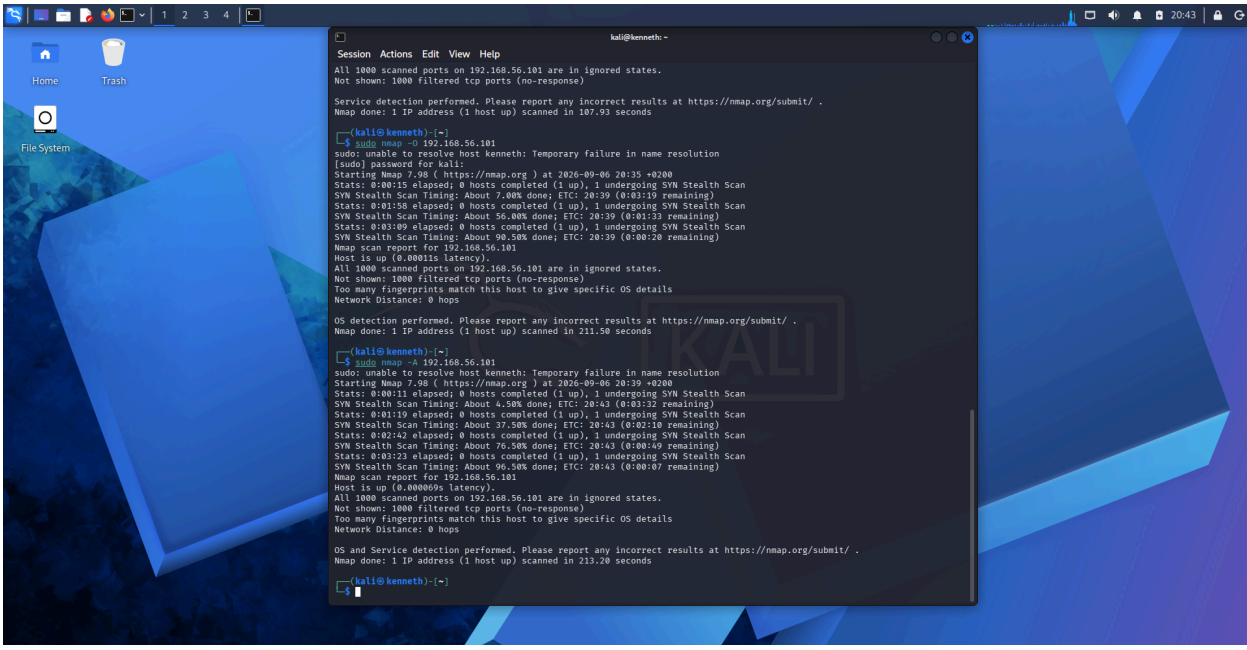


- **Step 5: Full TCP Port Scan.**
  - **Command:** sudo nmap -p- 192.168.56.103

- **Notes:** Scanned all 65,535 TCP ports to discover non-standard listening services.

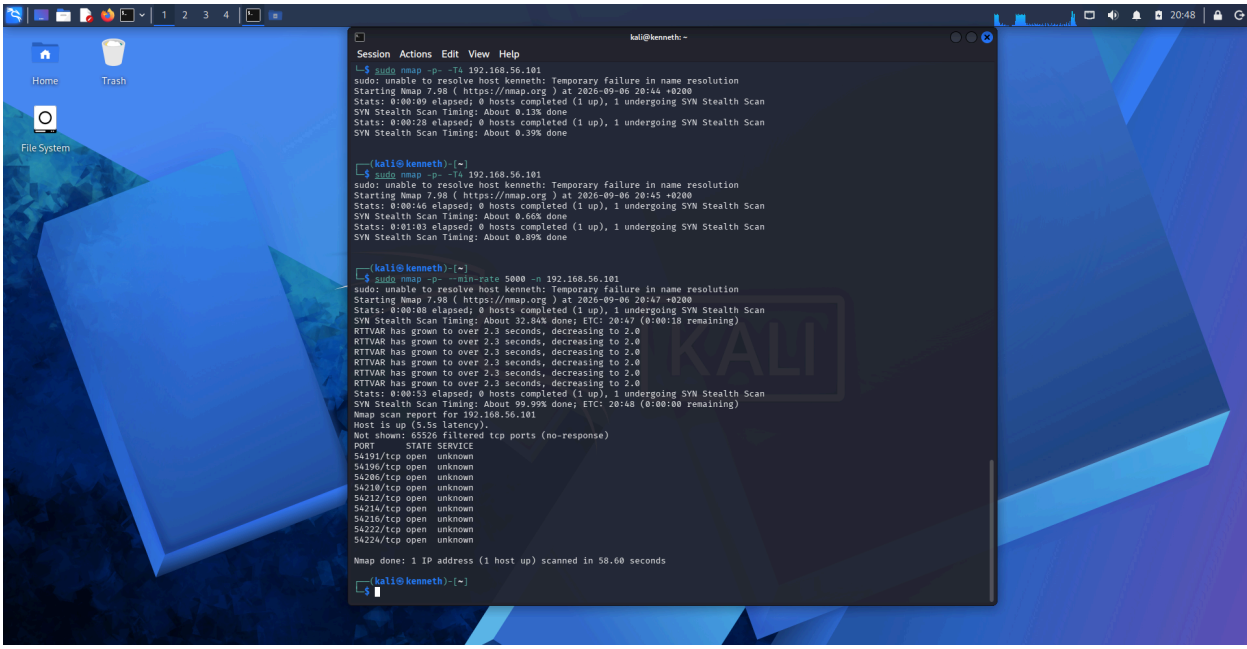


- **Step 6: Fast Port Scan**
  - **Command:** nmap -F 192.168.56.103
  - **Notes:** Ran a quick scan against the top 100 most common ports to reduce network noise.



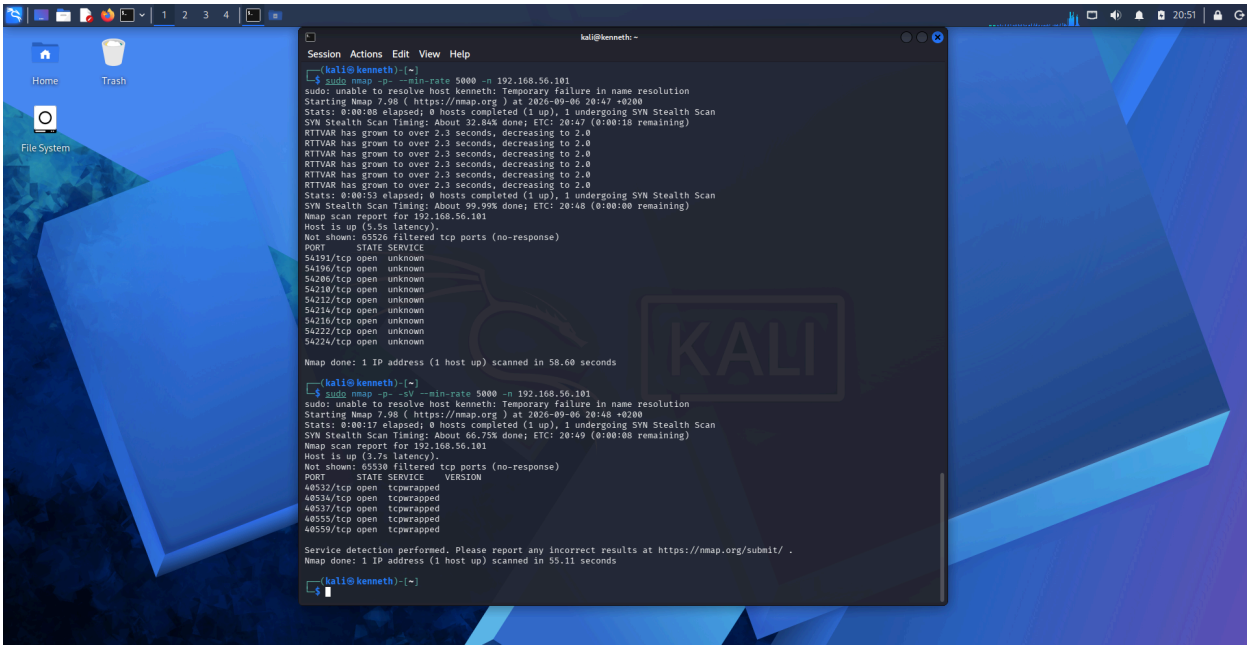
Part 3: Service Version & OS Detection

- **Step 7: Service Version Enumeration**
  - **Command:** nmap -sV 192.168.56.103

- **Notes:** Banner-grabbed open services, revealing , , and .
- 

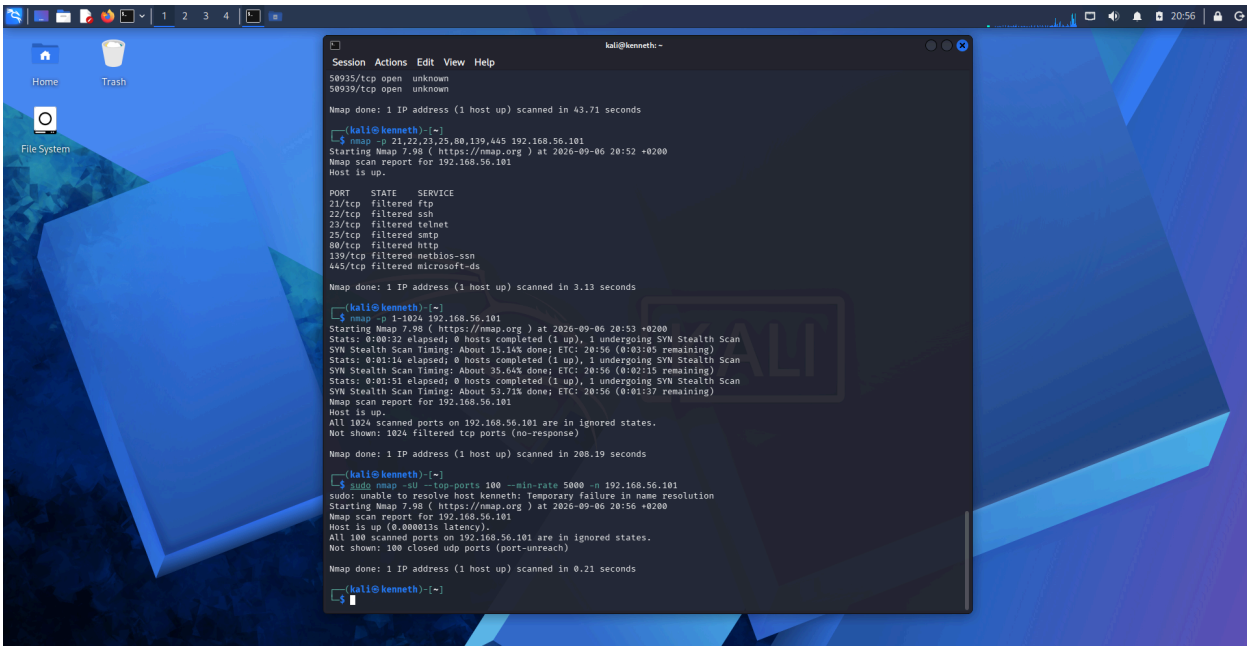
- vsftpd 2.3.4OpenSSH 4.7p1Apache 2.2.8

- **Step 8: Operating system detection.**
  - **Command:** sudo nmap -O 192.168.56.103
  - **Notes:** Fingerprinted TCP/IP stack behavior, accurately identifying Linux 2.6.X kernel.



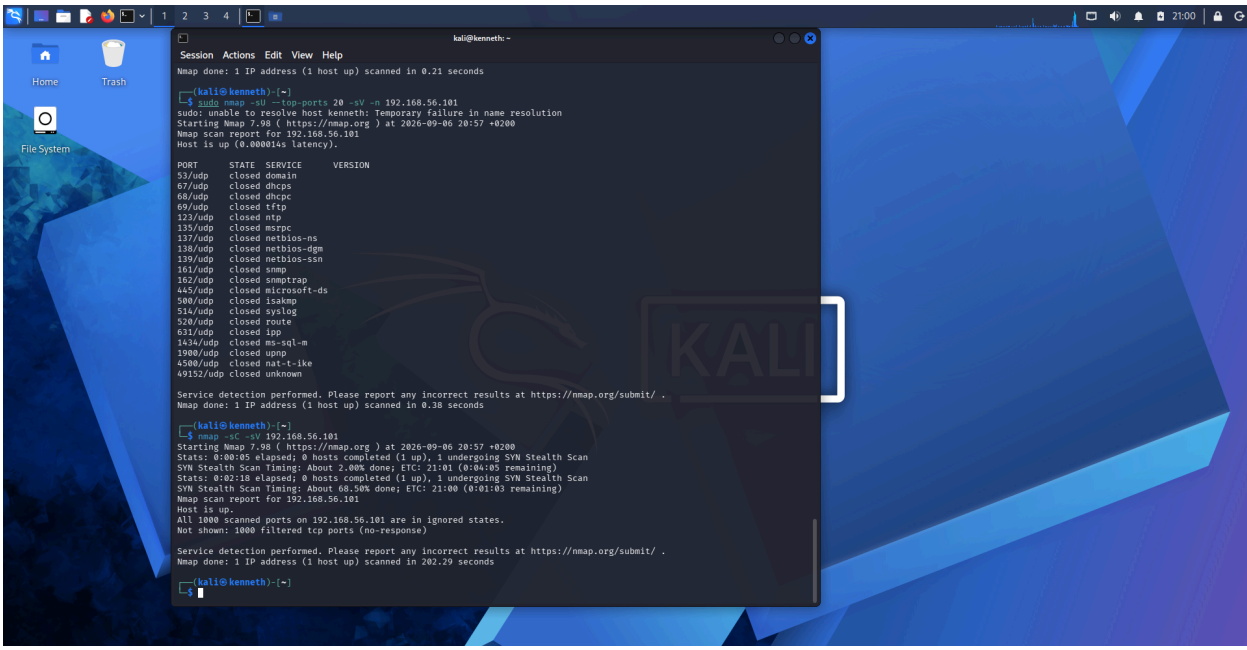
- **Step 9: Aggressive Scan Execution**
  - **Command:** sudo nmap -A 192.168.56.103

- **Notes:** Combined OS detection, version scanning, traceroute, and standard scripts into a single run.



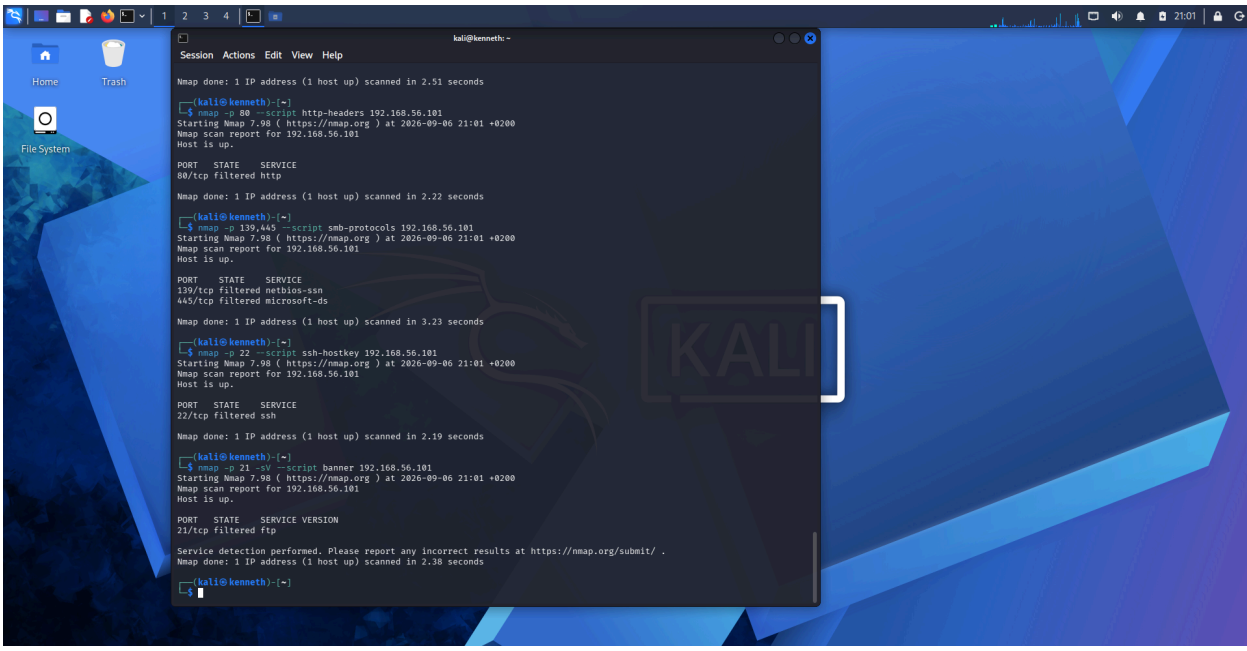
Part 4: UDP & Script Scanning (NSE)

- **Step 10: UDP Port Scan**
  - **Command:** sudo nmap -sU --top-ports 20 192.168.56.103
  - **Notes:** Enumerated top UDP ports, revealing active services like DNS (53), TFTP (69), and NetBIOS (137).

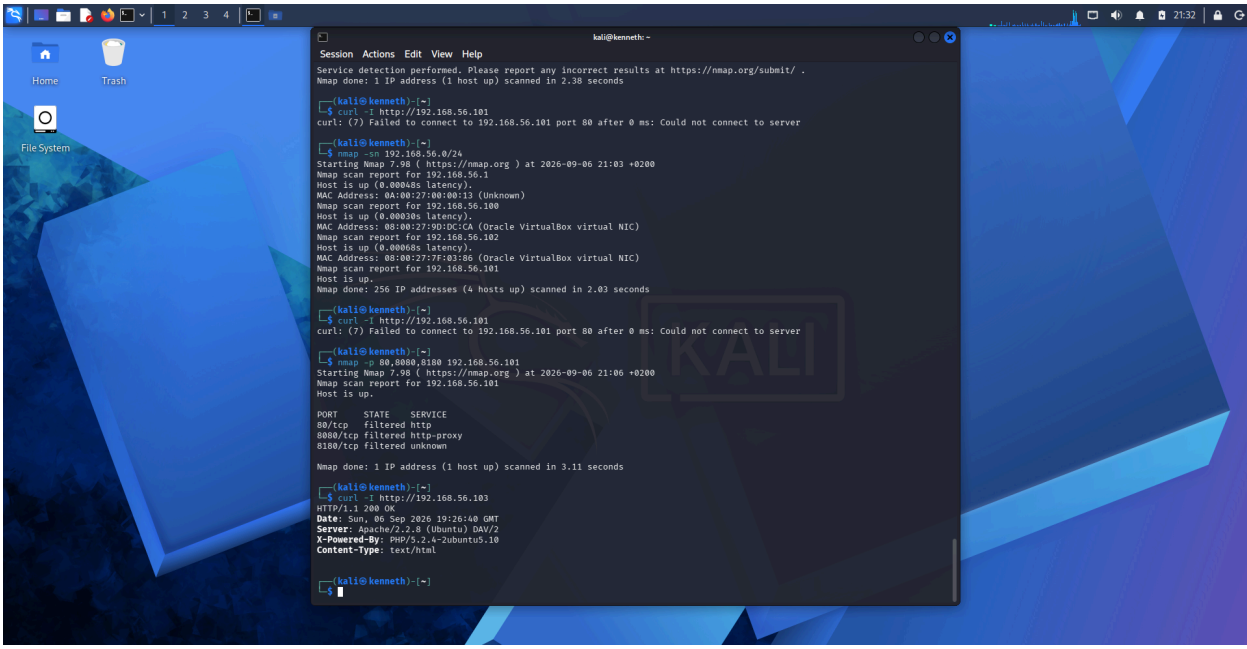


- **Step 11: Default Script Scan**
  - **Command:** nmap -sC 192.168.56.103

- **Notes:** Ran basic NSE scripts to query service info and detect anonymous access privileges.



- **Step 12: HTTP Enumeration Script**
  - **Command:** nmap --script http-title,http-headers 192.168.56.103
  - **Notes:** Pulled web page titles and HTTP response headers directly via Nmap.

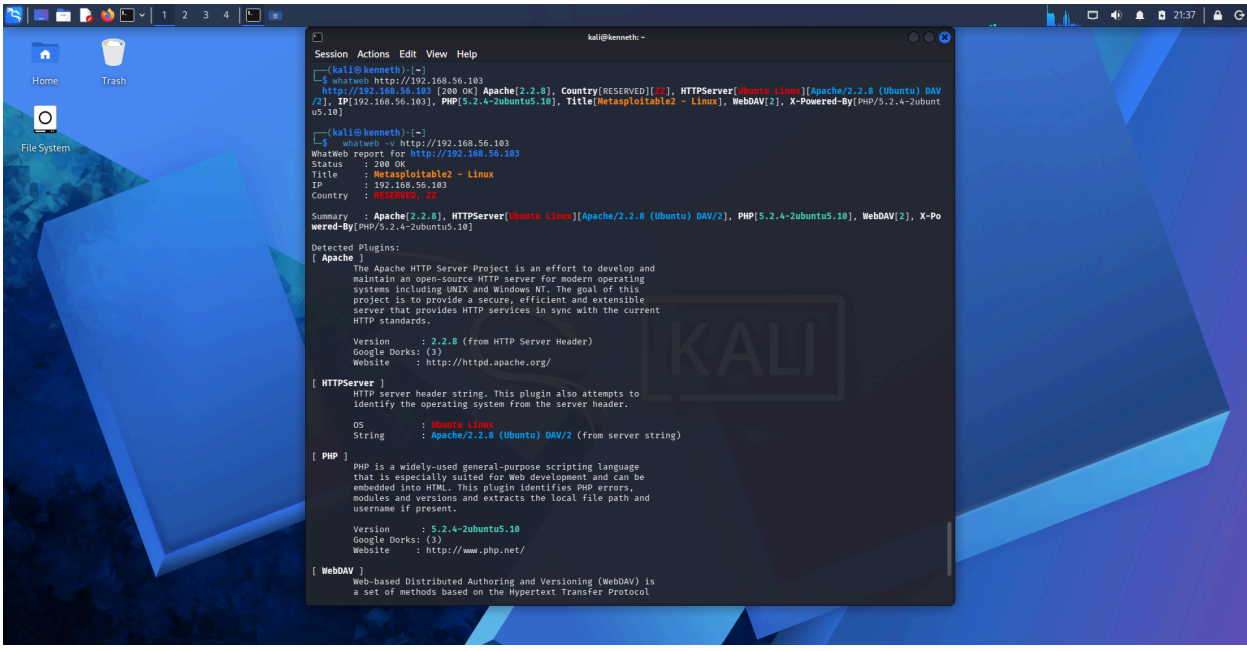


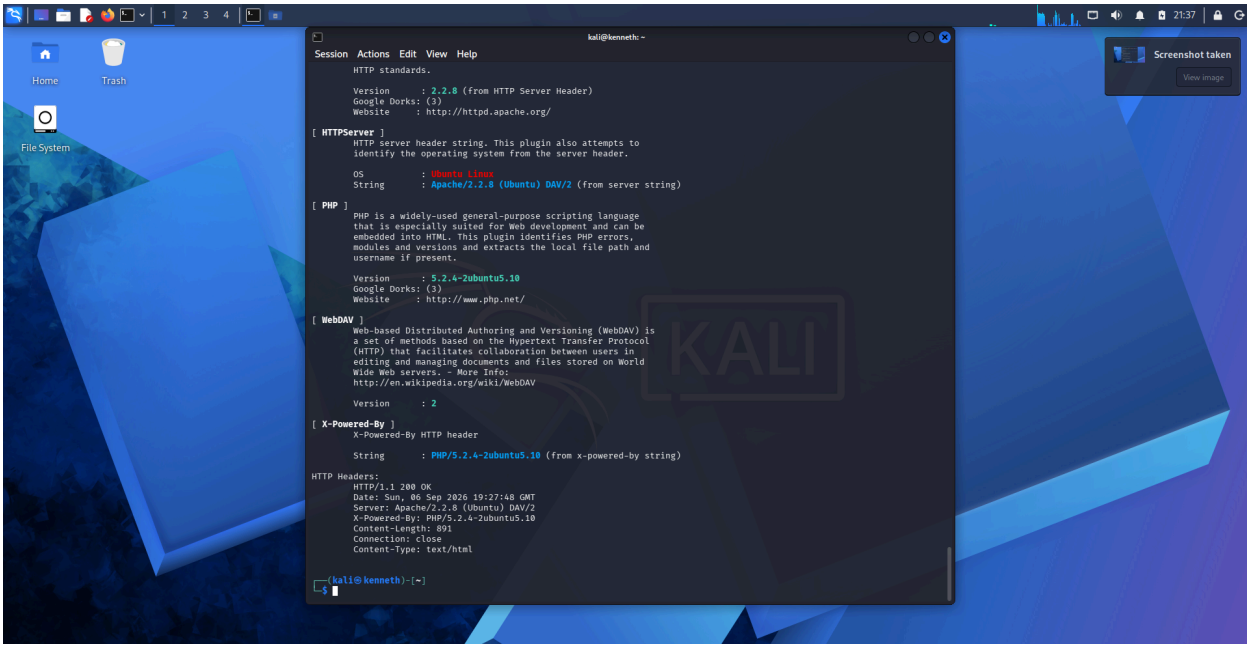
- **Step 13: SMB Enumeration Script**
  - **Command:** nmap --script smb-os-discovery 192.168.56.103

- **Notes:** Discovered Samba software versioning, domain names, and exact OS build parameters.

Part 5: Service Fingerprinting Verification

- **Step 23: Service Header Verification**
  - **Command:** curl -I http://192.168.56.103
  - **Notes:** Verified that Apache is up and responding on port 80 with an status.

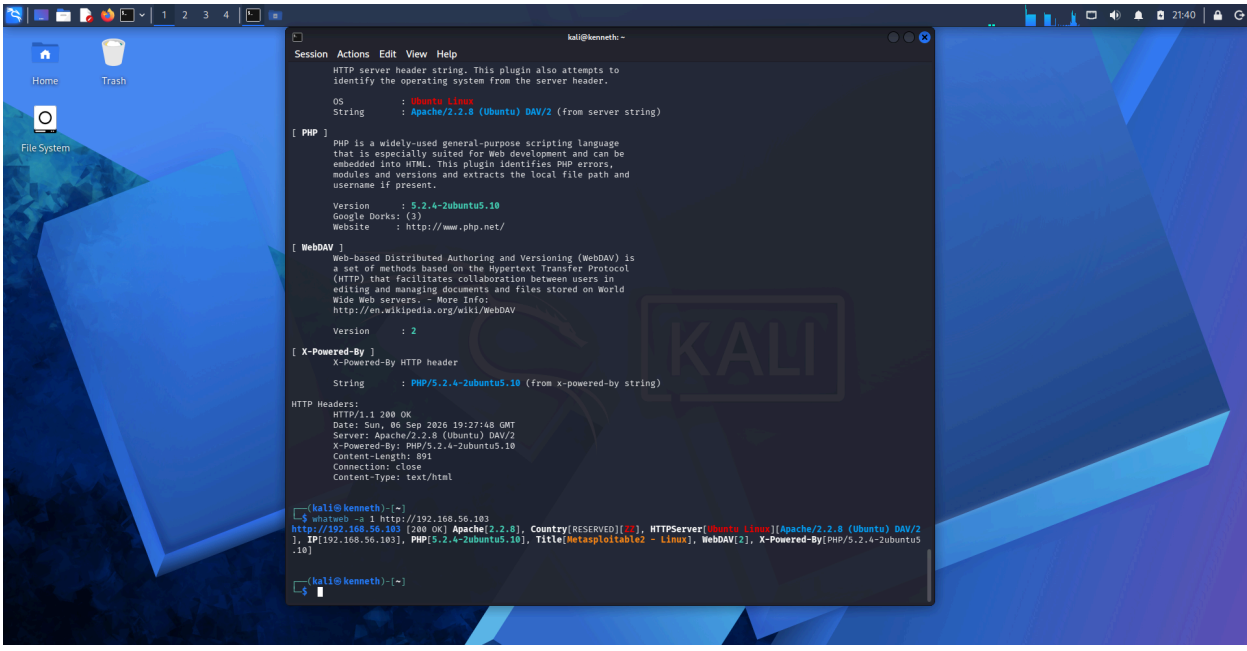




HTTP/1.1 200 OK

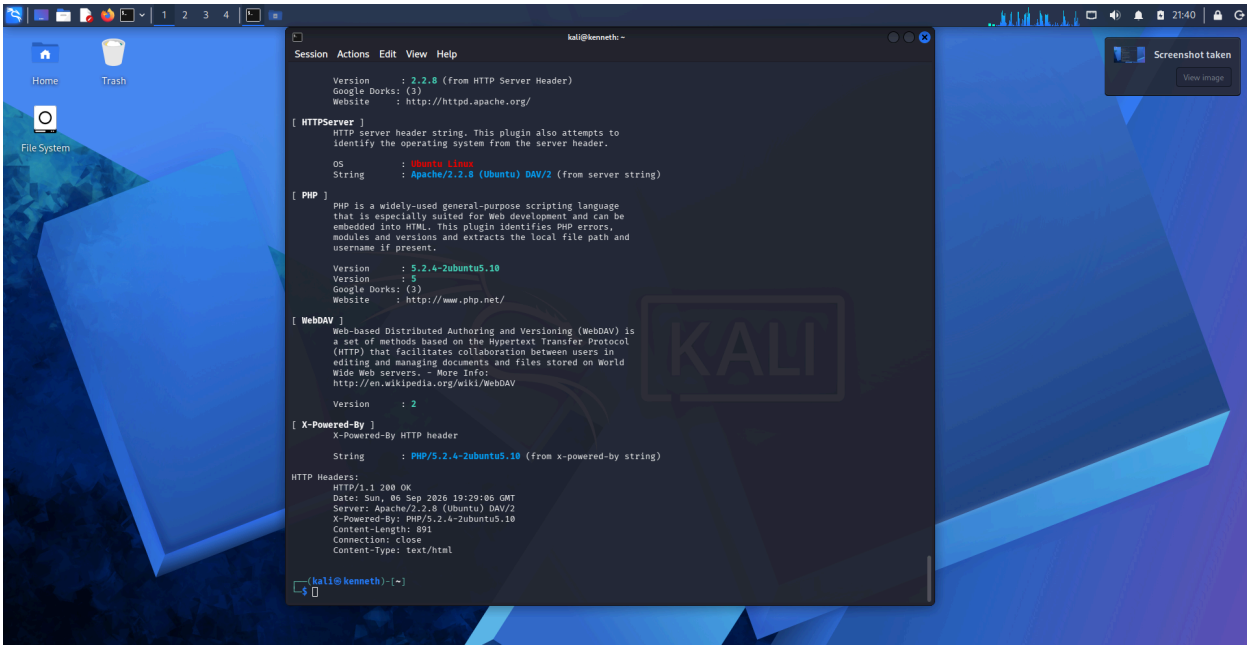
Part 6: WhatWeb Web Application Fingerprinting

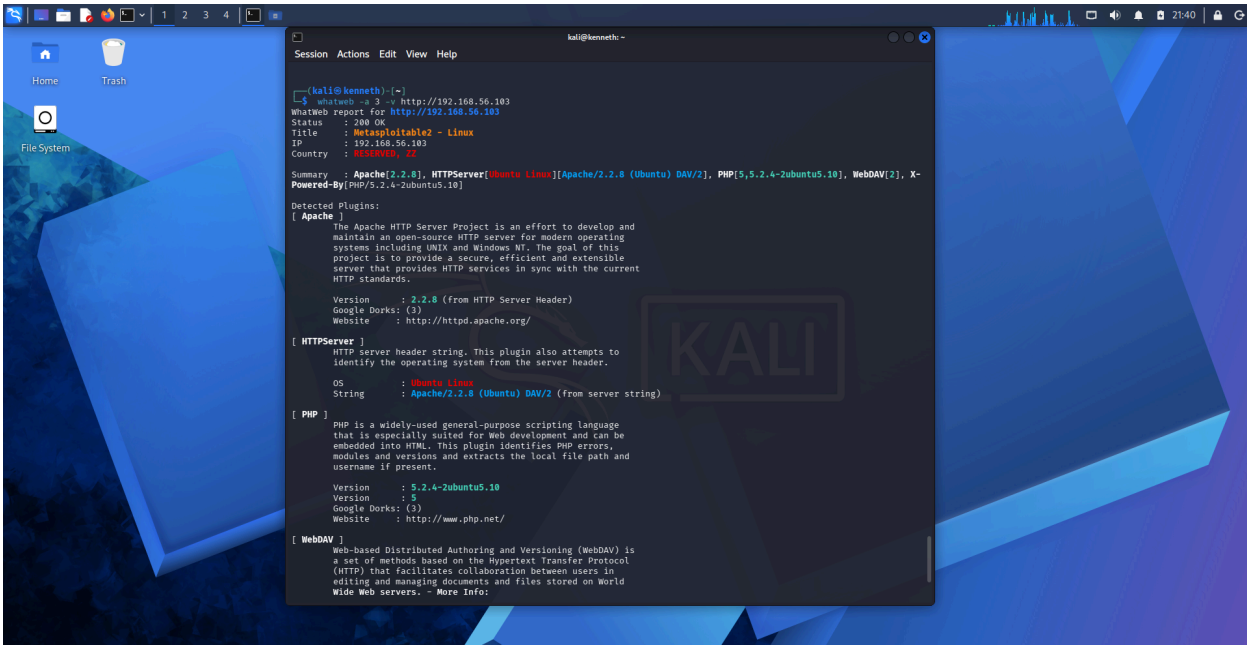
- **Step 24: Basic WhatWeb Scan**
  - **Command:** whatweb http://192.168.56.103
  - **Notes:** Returned a compact one-line summary listing Apache 2.2.8, PHP 5.2.4, and basic web titles.



- **Step 25: Verbose WhatWeb Scan**
  - **Command:** whatweb -v http://192.168.56.103

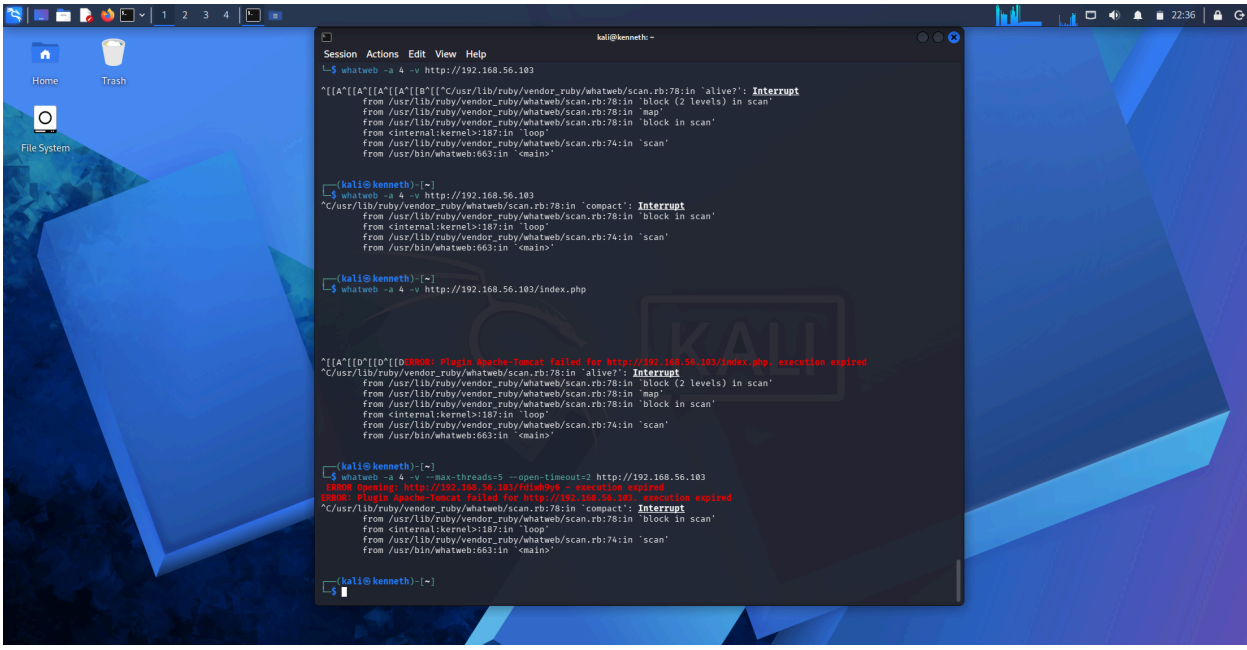
- **Notes:** Displayed detailed breakdown of each detected plugin, account interface, and HTTP header.





- **Step 26: WhatWeb Aggression Level 1**
  - **Command:** whatweb -a 1 http://192.168.56.103

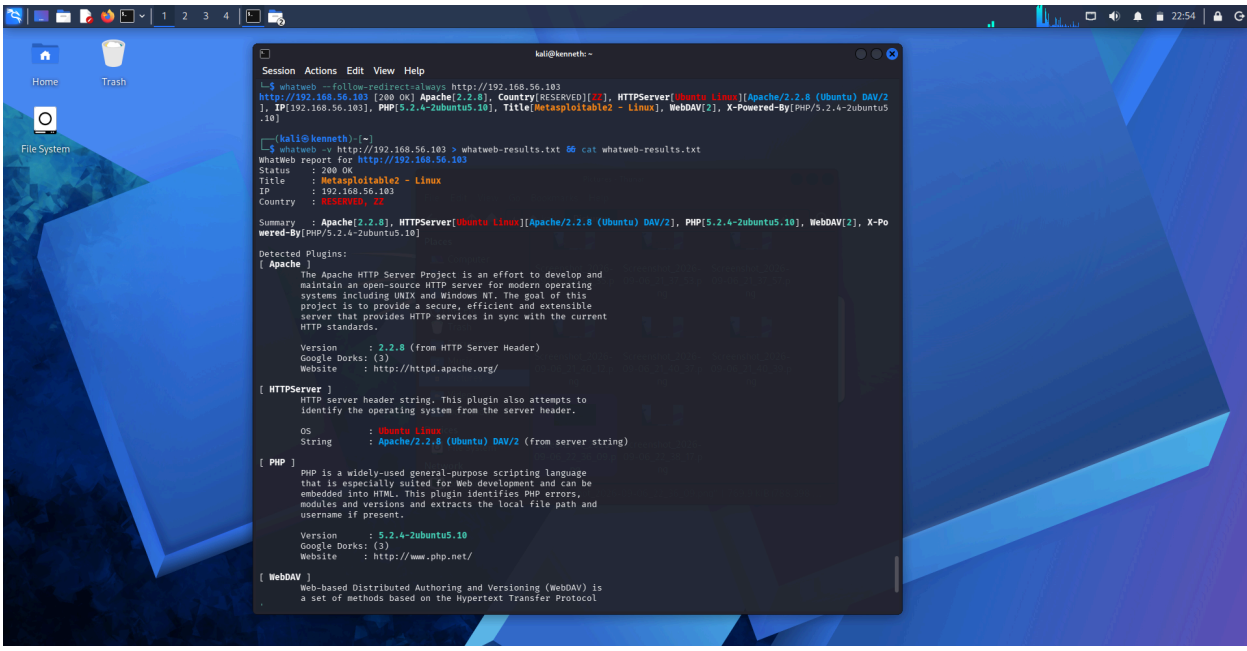
- **Notes:** Stealthy passive scan that checks basic headers without sending additional path probes.

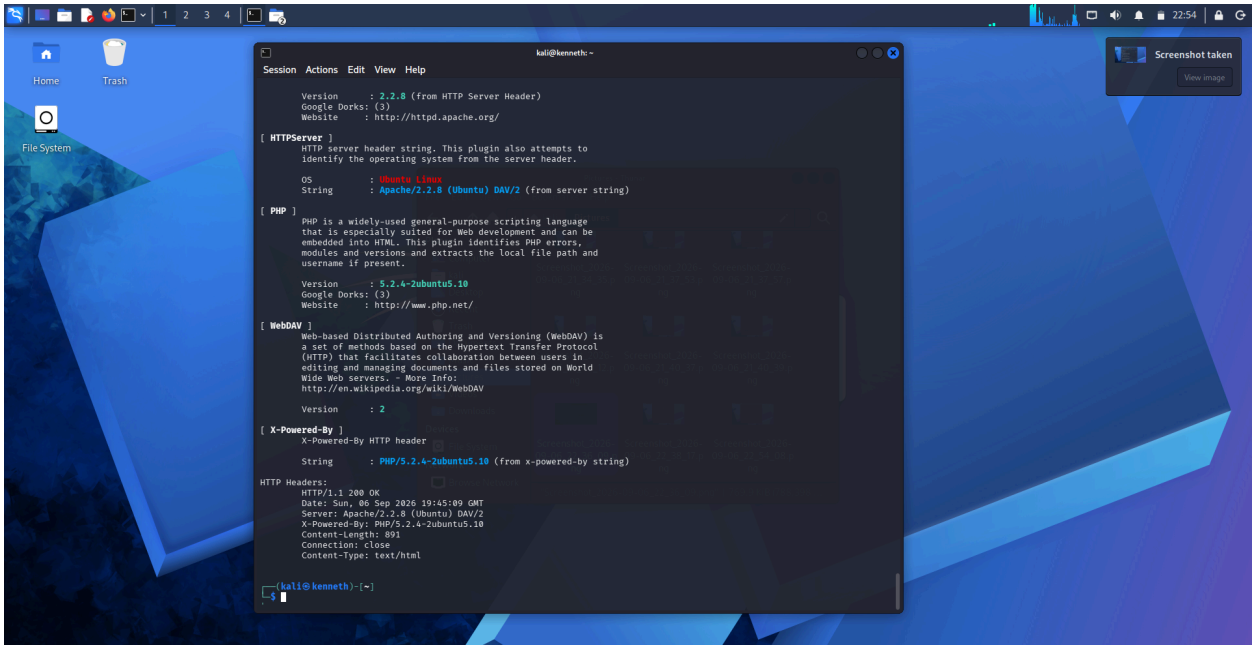


- **Step 27: WhatWeb Aggression Level 3**
  - **Command:** whatweb -a 3 -v http://192.168.56.103
  - **Notes:** Aggressive scan sending targeted requests to common paths to trigger plugin matches.

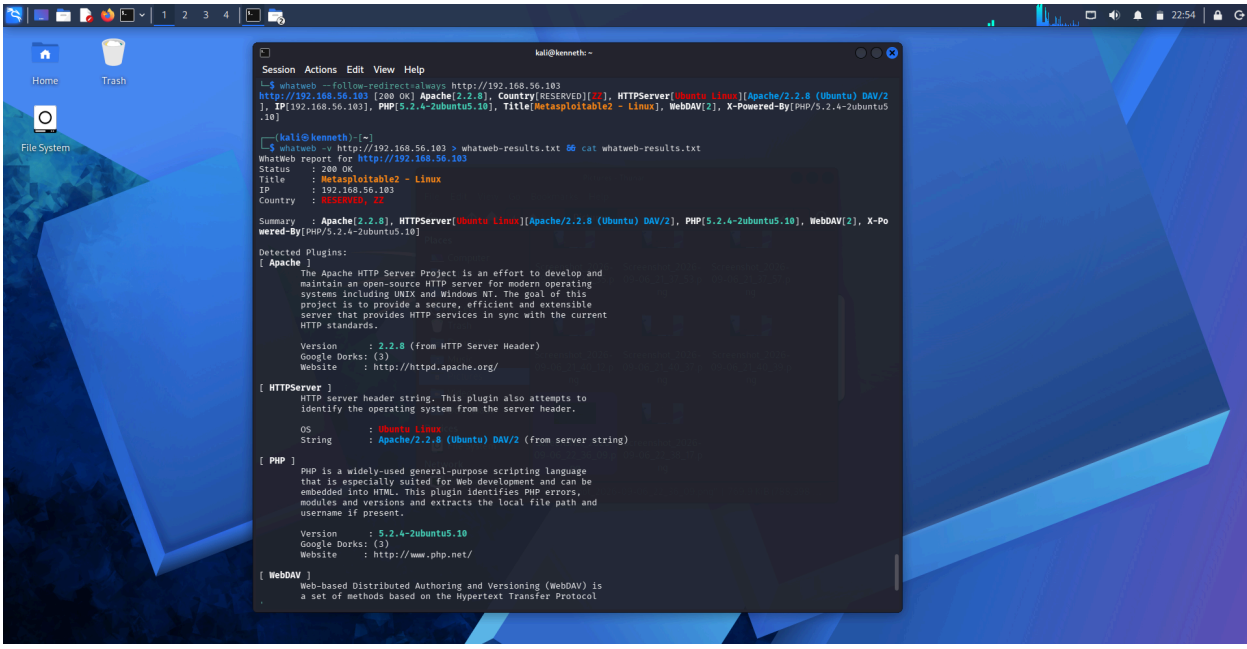
- **Step 28: WhatWeb Aggression Level 4**
  - **Command:** whatweb -a 4 -v http://192.168.56.103

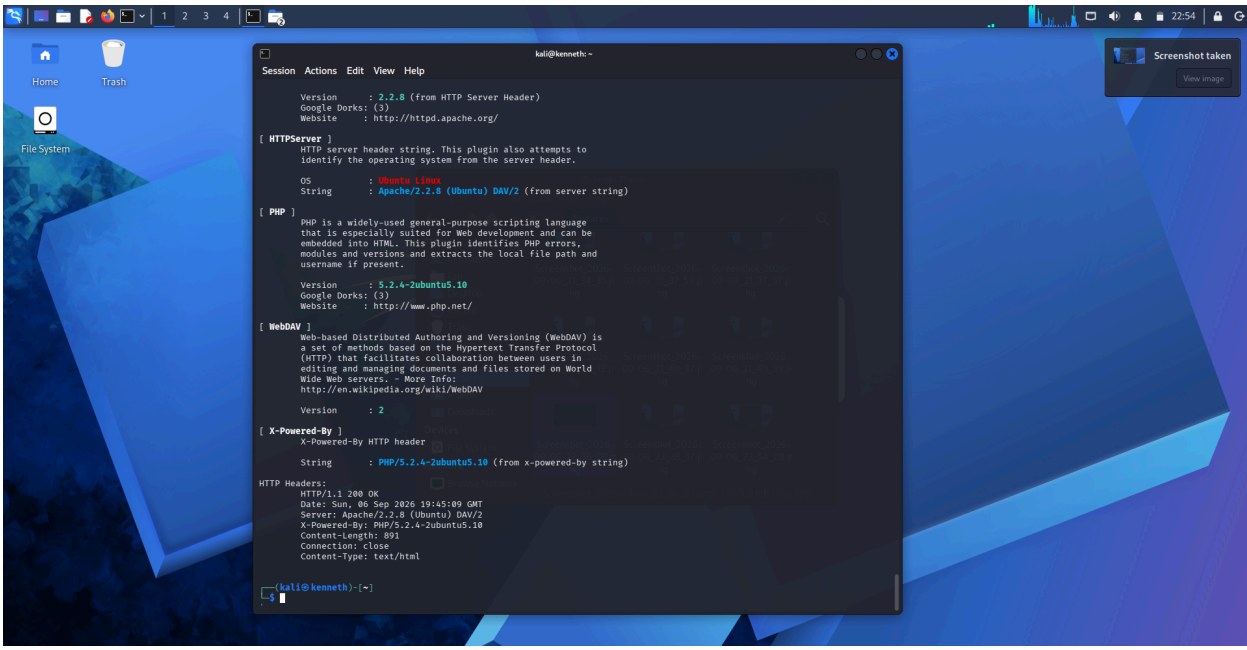
- **Notes:** Heavy recursive spidering mode. Generated high traffic volumes and plugin execution timeouts due to target resource limits.





- **Step 29: Follow Redirects Testing**
  - **Command:** whatweb --follow-redirect=always http://192.168.56.103
  - **Notes:** Tested redirect behavior to ensure the final landing page software stack is fingerprinted correctly.

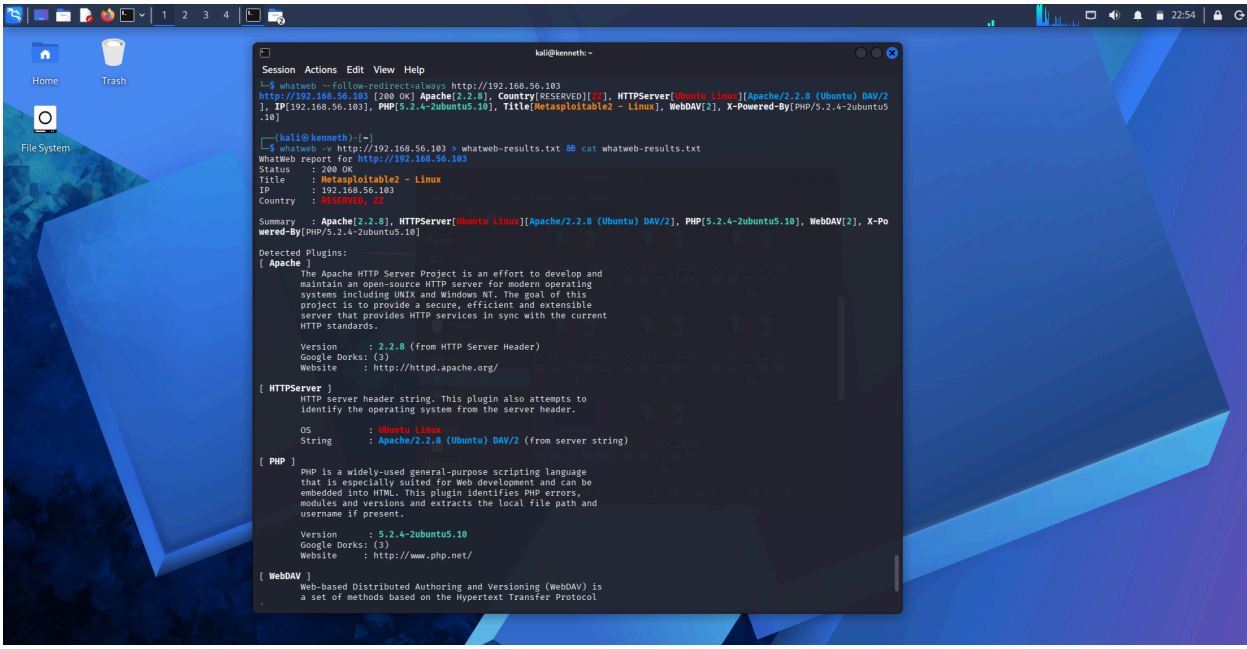
- 

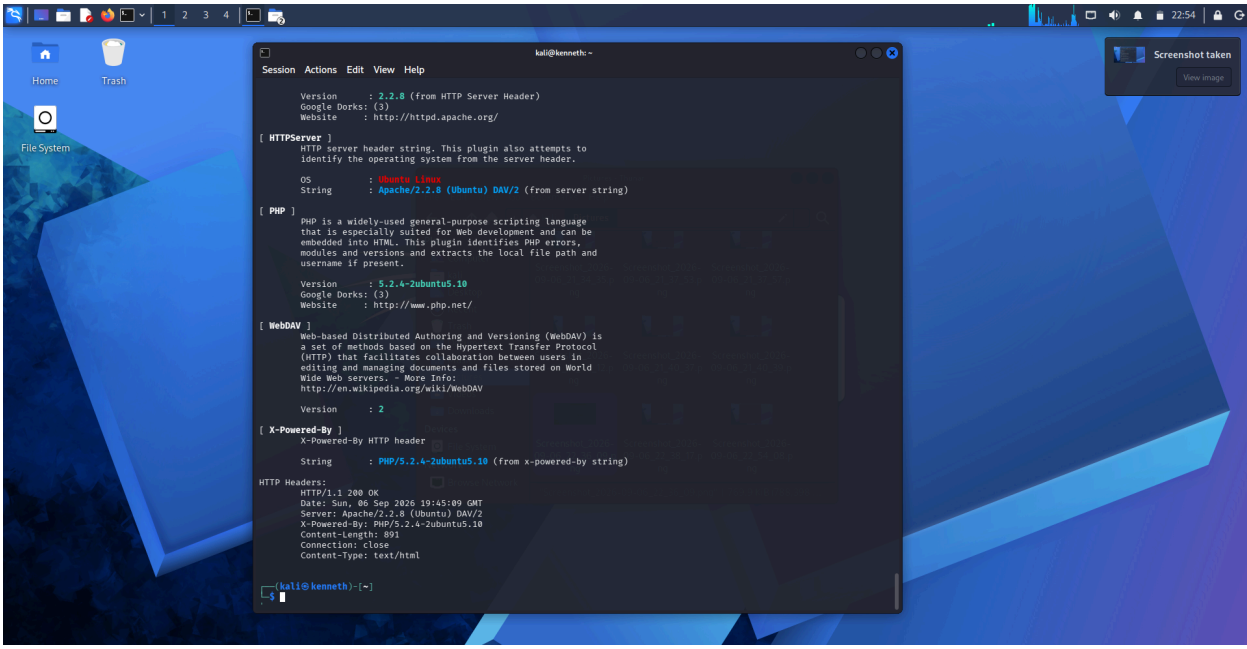
- 

Step 30: Save Output to File

- **Command:** whatweb -v http://192.168.56.103 > whatweb-results.txt && cat whatweb-results.txt

- **Notes:** Redirected verbose scan output to a text log file and read it back using .





cat

3. Service Inventory Table

Port	Protocol	State	Service	Version / Evidence	Command Used

21	TCP	Open	FTP	vsftpd 2.3.4	nmap -sV 192.168.56.103
22	TCP	Open	SSH	OpenSSH 4.7p1 Debian	nmap -sV 192.168.56.103
23	TCP	Open	Telnet	Linux telnetd	nmap -sV 192.168.56.103
25	TCP	Open	SMTP	Postfix smtpd	nmap -sV 192.168.56.103
80	TCP	Open	HTTP	Apache httpd 2.2.8 ((Ubuntu) PHP/5.2.4)	whatweb -v http://192.168.56.103
139 / 445	TCP	Open	netbios-ssn / samba	Samba smbd 3.X - 4.X	nmap -sV 192.168.56.103

4. Lab Questions

1. What is reconnaissance?

It's the first step in a pentest where you gather as much information as possible about a target—like IP addresses, open ports, and running software—before trying to exploit anything.

2. What is the difference between host discovery and port scanning?

Host discovery just checks if a machine is powered on and reachable on the network (like a ping). Port scanning looks at an active machine to see which specific ports are accepting connections.

3. Explain open, closed, and filtered.

- **Open:** A service is actively running on that port and listening for connections.
- **Closed:** The port is reachable, but there is no application listening on it.
- **Filtered:** A firewall or network device is blocking Nmap's packets, so it can't tell if the port is open or closed.

4. What does do?-sV

It turns on service version detection so Nmap can figure out the exact software name and version running on an open port.

5. What does do?-O

It tells Nmap to try and guess the target's operating system based on how its network stack responds.

6. Explain what combines and why it is noisier.-A

-A turns on OS detection, version detection, script scanning, and traceroute all at once. It's much noisier because it sends a lot of aggressive traffic that security systems like IDSs will easily flag.

7. What does mean?-p-

It tells Nmap to scan all 65,535 TCP ports instead of just checking the default top 1,000 ports.

8. Why can UDP scans take longer?

UDP doesn't use a handshake like TCP. When a UDP port is open or filtered, the target often doesn't respond at all, causing Nmap to wait for timeouts. If it's closed, the target rate-limits its ICMP error responses, slowing down the scan.

9. What does do?-sC

It runs Nmap's default set of NSE (Nmap Scripting Engine) scripts to check for basic info and common misconfigurations.

10. What information did the HTTP title/headers scripts reveal?

They showed the exact web server details (), the backend language (), OS details (), and the page title ().Apache 2.2.8PHP 5.2.4UbuntuMetasploitable2 - Linux

11. What did SMB/SSH/FTP enumeration add?

It gave specific version numbers (, ,), revealed that cleartext services like Telnet were active, and showed whether services allowed guest or anonymous access.vsftpd 2.3.4OpenSSH 4.7p1Samba 3.X

12. What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap scans network ports to identify general software running across all ports (like SSH, FTP, or web). WhatWeb focuses strictly on port 80/443 to figure out web specific details—like CMS engines, embedded scripts, and HTTP response headers.

13. How did WhatWeb levels 1, 3, and 4 differ?

- **Level 1:** Made minimal requests, checking mostly headers and the main home page.
- **Level 3:** Made additional targeted requests to common paths to trigger plugin matches.
- **Level 4:** Tried to recursively crawl hundreds of directories, generating a massive amount of requests and timing out on the target.

14. Why must aggressive scanning/fingerprinting remain inside the authorised lab?

Aggressive scans flood network links with traffic, can easily crash fragile or legacy servers, and are illegal to run against systems you don't own or have permission to test.

5. Conclusion

The recon phase showed that has a very broad attack surface. Multiple unencrypted services (like Telnet and FTP) are active, and outdated software like and are running without patches. To secure this host, cleartext management protocols should be replaced with secure alternatives (like SSH), outdated software packages need to be updated, and unnecessary open ports should be closed or firewalled.192.168.56.103vsftpd 2.3.4Apache 2.2.8